Md Ahsanul Haque, Ph.D.
MAT-120

Quiz-2

Date: July, 2026

Name

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1	10	
2	10	
Total	20	

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation and show all logical steps.
- Check that your answer can be read and understood by the grader.

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1. Consider the function

$$f(x) = \begin{cases} \dfrac{e^{2x} - e^x}{e^x - 1}, & x \neq 0, \\ k, & x = 0. \end{cases}$$

   (a) Find the value of k that makes f continuous at $x = 0$.

   (b) State the type of discontinuity at $x = 0$ if $k \neq \lim_{x \to 0} f(x)$. Justify your answer algebraically.

2. (a) Let $g(x) = \sqrt{x+2} - x^2$. Determine if the equation $g(x)$ has at least one real solution in the interval $[1, 2]$.

   (b) Let

$$p(x) = \begin{cases} 2 - x^2, & x \leq 1, \\ \dfrac{2}{x+1}, & x > 1. \end{cases}$$

   Determine whether $p(x)$ is differentiable at $x = 1$. Show all necessary steps.